

Unit 13, Lesson 1: Angle Relationships

Benchmark

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.

Learning Target

☐ I can identify adjacent, supplementary, complementary, and vertical angles.

13.1.1 Warm-Up: Sandy Beaches

There are so many beaches around the world that most people can't tell whether they are natural or man-made. For example, the 7.5-acre Crystal Lagoon in Wesley Chapel, Florida includes a man-made beach.



1. To create a man-made beach, sand needs to be brought in to construct it. One cubic foot of sand weighs around 100 pounds. One acre covers about 43,560 square feet.

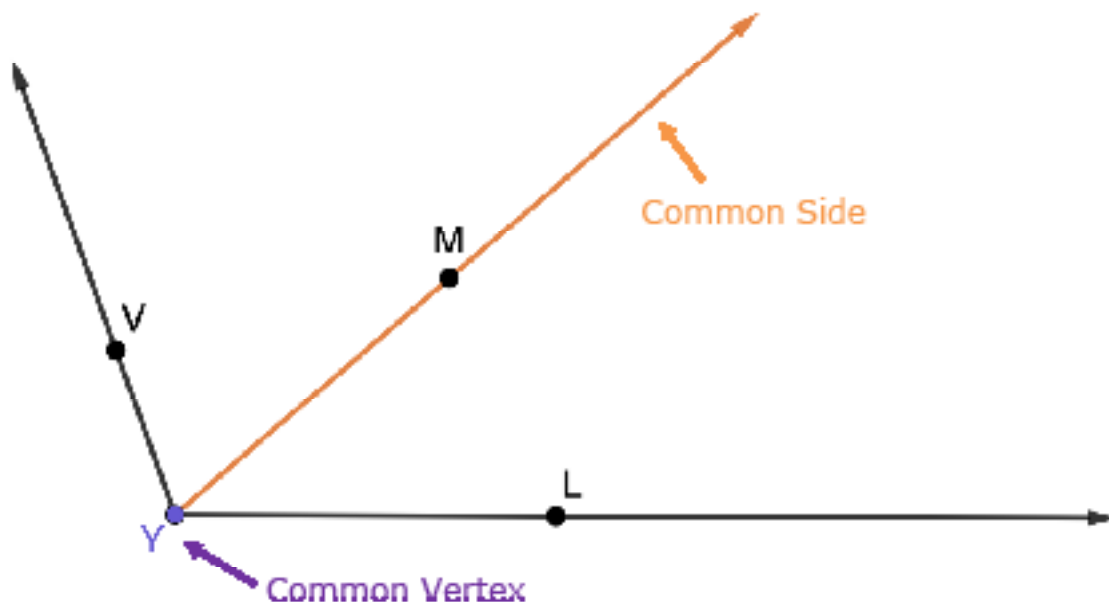
What is the weight of the sand that needs to be brought in to cover a 1-acre area with 3 inches, or $\frac{1}{4}$ foot, of sand?

13.1.2 Guided Instruction: Adjacent Angles

Angles are formed when lines, line segments, or rays intersect.

Two angles that share a side and vertex and do not overlap are called **adjacent angles**.

The figure shows an example of adjacent angles $\angle VYM$ and $\angle MYL$.

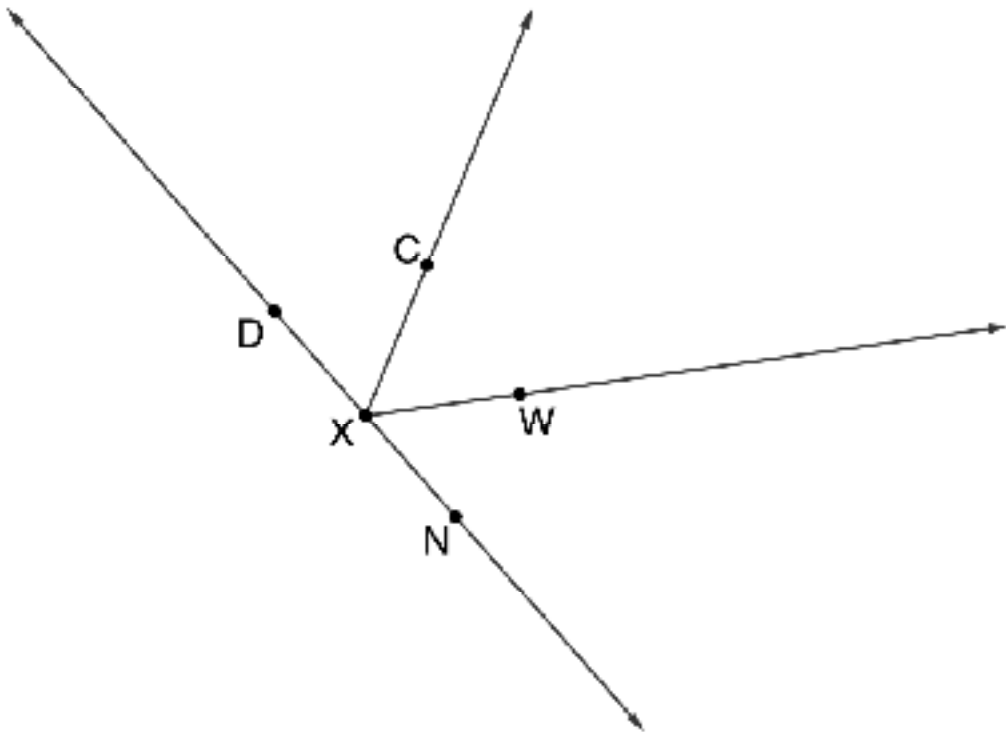


Point Y is the common vertex.

Ray $\overrightarrow{YM}$ is the common side that both angles share.

1. The measure of $\angle VYM$ is 68° and the measure of $\angle MYL$ is 41° .
 - a. What is the measure of $\angle VYL$?
 - b. Discuss with your partner how you know the measure of $\angle VYL$ from the definition of adjacent angles and the given information.

2. Line DN , ray XC , and ray XW intersect to form angles, as shown.



Complete the table to identify whether each pair of angles given are adjacent angles.

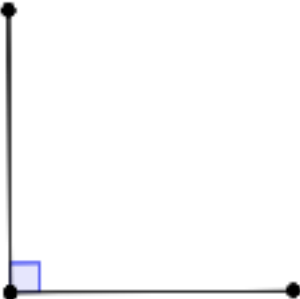
Angle Pair	Angle Relationship
$\angle CXD$ and $\angle CXW$	☐ adjacent angles ☐ not adjacent angles
$\angle CXW$ and $\angle DXW$	☐ adjacent angles ☐ not adjacent angles
$\angle DXW$ and $\angle WXN$	☐ adjacent angles ☐ not adjacent angles
$\angle NXC$ and $\angle DXC$	☐ adjacent angles ☐ not adjacent angles
$\angle DXC$ and $\angle NXW$	☐ adjacent angles ☐ not adjacent angles

13.1.3 Exploration: Angle Relationships

1. Several angle names and measures are given in the bank below.

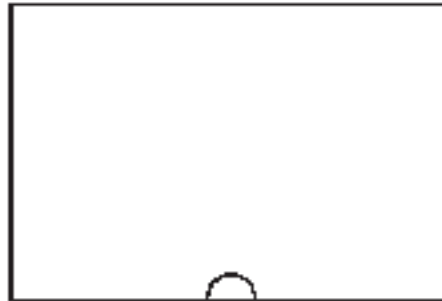
acute angle	obtuse angle	20°	45°	90°
right angle	straight angle	125°	180°	360°

Work with your partner to complete the table by matching the angle name and measure from the bank to each type of angle shown. Some words and angle measures from the bank will not be used.

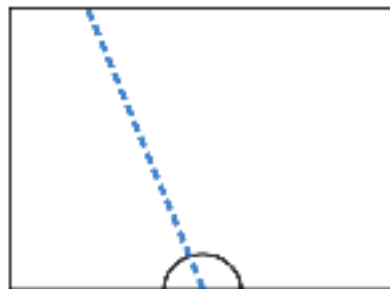
Image	Angle Name	Angle Measure
		
		

Your teacher will give you three rectangular pieces of paper to use for the remainder of the Exploration.

2. Use one rectangular piece of paper to complete the following.
- a. Draw a small half-circle in the middle of one side, as shown.



- b. Use a straightedge to draw a line starting from the center of the half-circle, all the way across the paper. The line can go in any direction or angle across the paper. Two possible examples are shown.



- c. Cut your paper along the line you drew.
- d. On each of the two pieces created, use a protractor to measure the angle that is marked by part of the circle. Then, label the angle measure on the paper.
- e. Write two observations about the two angles.
- f. Clare measured the angle on one of her pieces to be 70° . Discuss with your partner how to predict the angle measure of her other piece. Then, write your prediction.

3. Use a second rectangular piece of paper to complete the following.

a. Draw a small quarter-circle in one of the corners.



b. Use a straightedge to draw a line starting from the vertex with the quarter-circle, all the way across the paper. The line can go in any direction or angle across the paper.

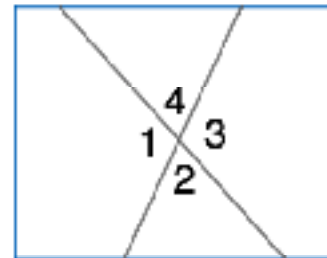
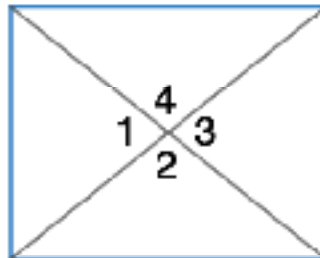
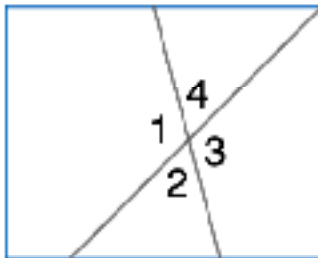
c. Cut your paper along the line you drew.

d. On each of the two pieces created, use a protractor to measure the angle that is marked by part of the circle. Then, label the angle measure on the paper.

e. Add your two angle measures together. What do you notice?

f. Priyanka measured the angle of one of her pieces to be 53° . Discuss with your partner how to predict the angle measure of her other piece. Then, write your prediction.

4. Use the third rectangular piece of paper to complete the following.
- a. Use a straightedge to draw two intersecting lines across the paper. Then, label the angles created with 1, 2, 3, and 4. Three possible examples are shown.



- b. Cut across the straight lines you drew, creating four separate pieces.
- c. Rotate and flip the angles to determine how they compare with each other. This could include laying angles on top of one another to compare their size. Summarize your observations of how the angles compare to each other.
- d. Compare your findings with your partner. Make a conjecture about the relationship between angle measures when two lines intersect.

13.1.4 Bring It Together: Angle Relationships

Three different angle relationships were explored in the Exploration. Their definitions are shown.



Two angles whose measures add up to 90° are *complementary angles*.



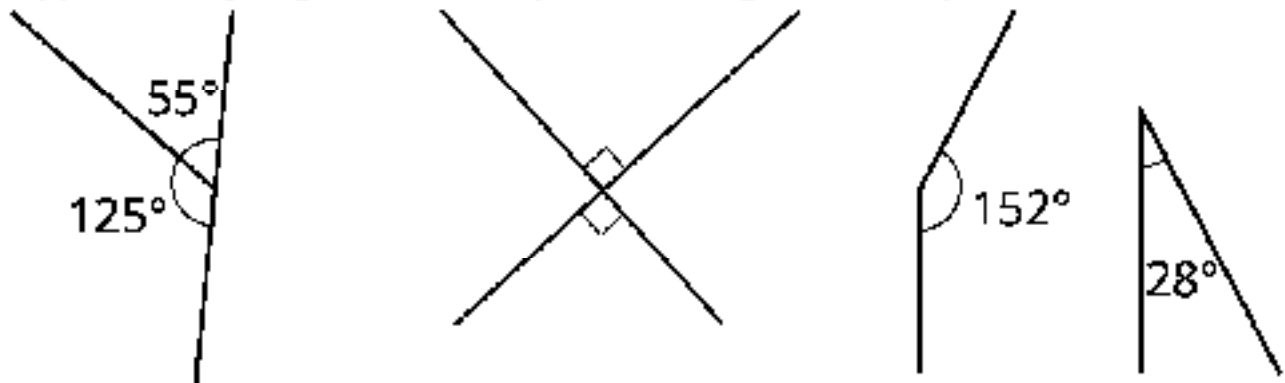
Two angles with measures that add up to exactly 180° are *supplementary angles*.



Vertical angles are the opposite angles formed when two lines intersect.

1. Complete the statements using the definitions of each angle relationship.
 - a. The angles created in question 2 of the Exploration are _____ angles.
 - b. The angles created in question 3 of the Exploration are _____ angles.
 - c. The angles created in question 4 of the Exploration are _____ angles.

2. **Supplementary angles can be adjacent or non-adjacent. Three examples of supplementary angles are shown, with the angles in each pair marked.**



Complete the statement.

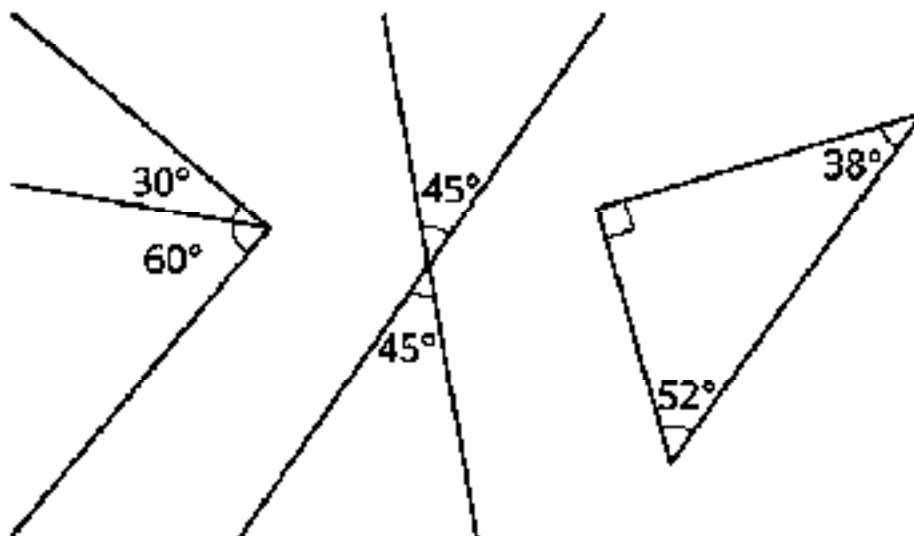
When supplementary angles are adjacent, they form

- ☐ an acute
- ☐ a right
- ☐ an obtuse
- ☐ a straight

angle together.

3. **Three examples of complementary angles are shown. Complementary angles can be adjacent or non-adjacent.**

- a. **Circle the example that shows an adjacent pair of complementary angles.**



b. Complete the statement.

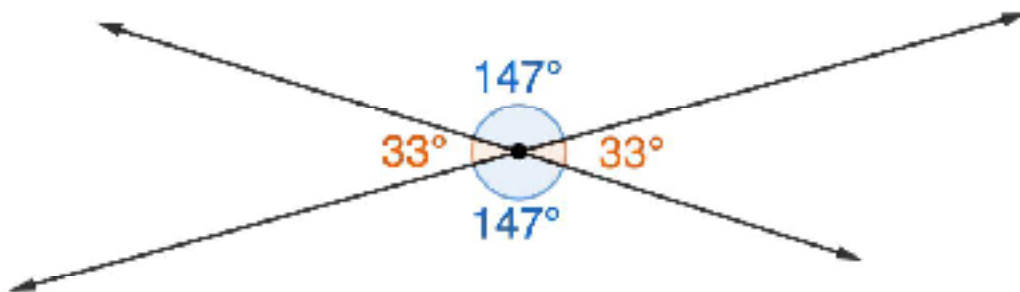
When complementary angles are adjacent, they

form

- ☐ an acute
- ☐ a right
- ☐ an obtuse
- ☐ a straight

angle together.

4. Each pair of vertical angles formed by the intersecting lines below are shown in a different color.



- a. Discuss with your partner the angle relationship between adjacent angles when two lines intersect.

b. Complete the statements.

Vertical angles

- ☐ sometimes
- ☐ always

have equal measures. Pairs of adjacent

angles created when two lines intersect are

- ☐ complementary.
- ☐ supplementary.

13.1.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...